ENGINEERING HANDBOOK | LIVING EDITION

AutoTune AI

An Agentic, Self-Healing MLOps Platform

Architecture, training diagnostics, guarded hyperparameter recovery, Kubeflow orchestration, Meta advisor integration, implementation roadmap and verified baseline.

Version: v0.7 | Checkpoint: Phase 5A Windows verified + 5B offline contracts | 01 November 2026

This handbook distinguishes CPU synthetic recovery from genuine small-model PyTorch training. No hosted Meta API, real Meta Llama GPU fine-tuning, or production deployment is claimed.

Executive overview

AutoTune AI is a constrained feedback controller for ML training and fine-tuning. Instead of relying on a human to constantly inspect loss curves, it observes training events, detects potential generalization or numerical issues, preserves a known-good candidate and starts a new **bounded** experiment with verified parameters. A Meta model may advise the diagnosis and hyperparameters, but it cannot directly execute code or rewrite arbitrary files.

Design promise: *autonomous experiments; deterministic guardrails; auditable evidence; explicit release control.*

> **Current truth at v0.7:** The user verified Phase 5A real PyTorch tiny-GRU training and 95 passing tests on Windows. New Phase 5B offline contracts, LoRA source pipeline and checkpoint manifest tests raise build-container coverage to 134 passing tests. Hosted Meta API, live Meta Llama GPU training, real Hugging Face optimizer resume, Kubeflow and remote MLflow remain unverified.

Core outcomes

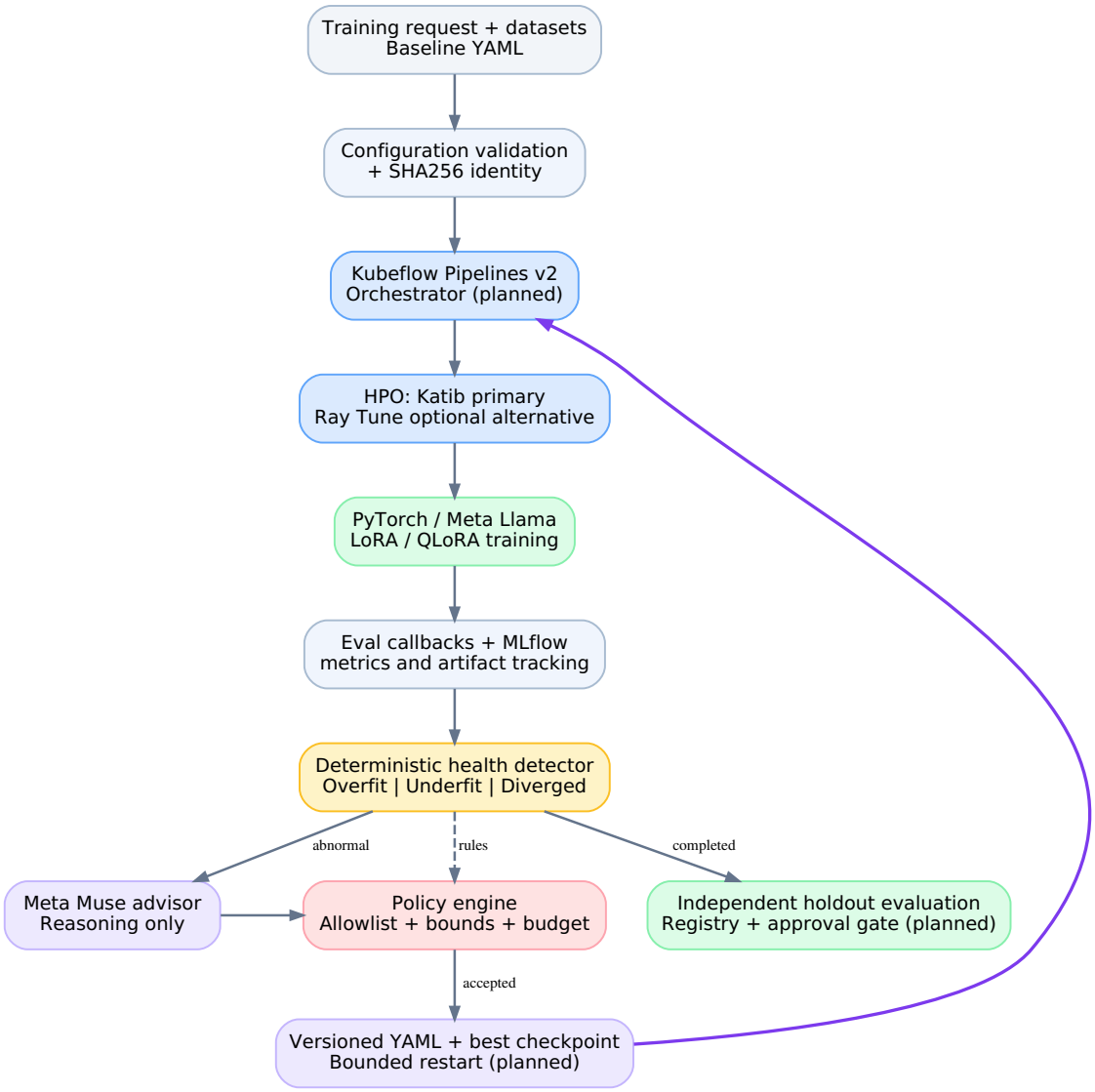
- Detect likely overfitting, underfitting and non-finite loss during repeated evaluations.
- Capture the best validation checkpoint **from each attempt**, with clear artifact identity.
- Generate small remediation proposals, validate them against strict numeric limits, and rerun only within configured budgets.
- Preserve baseline configurations, decision logs and SHA-256 configuration identity.
- Separate **optimization** (validation metrics) from **final release acceptance** (held-out evaluation, policy, approval).

Read this alongside

- `docs/MASTER_EXECUTION_PLAN.md` - all milestones, interfaces and exit criteria.
- `AGENTS.md` - compact history, decisions, pending items, verified checkpoints.

- `docs/diagrams/*.mmd` - editable Mermaid architecture and logic sources.
- `autotune/` and `tests/` - executable source and test evidence.

1. System architecture



End-to-end architectural components and the feedback loop

Interface explanation. A validated request moves from the configuration layer to an orchestrator. A training job emits evaluation metrics and writes checkpoints. A deterministic health analyzer determines whether to continue or stop. On a detected problem, the Meta agent optionally proposes a bounded patch; the policy engine decides whether to accept it. An approved patch creates a new

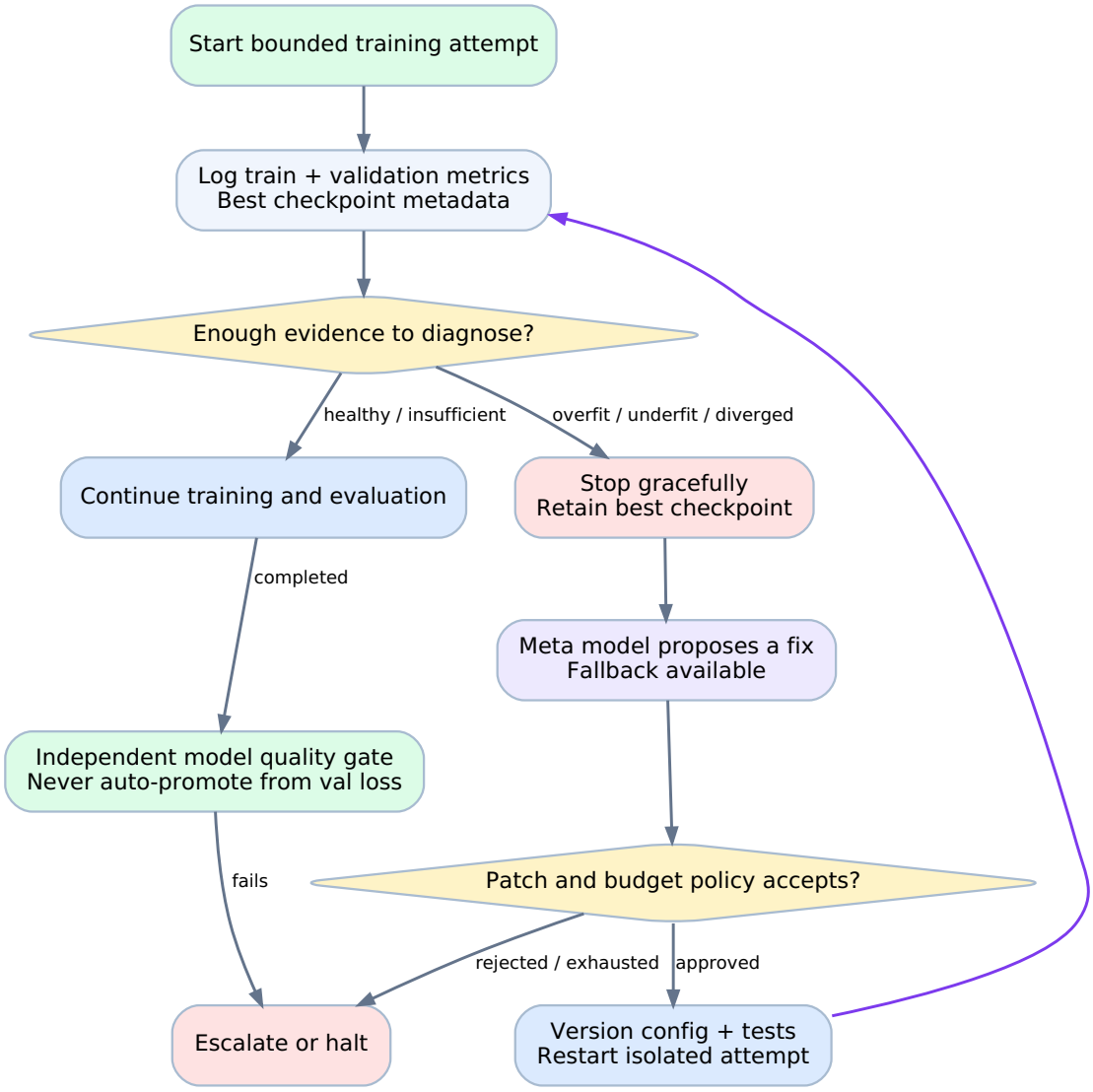
versioned configuration and a new training attempt. Completion does not imply deployment: the model must pass independent release evaluation.

Layer	Primary component	Responsibility	v0.7 status
Control plane	Python CLI / later FastAPI dashboard	Training request, configuration, run controls	Local CLI
Orchestration	Kubeflow Pipelines v2	Pipeline tasks and retry scheduling	Example only
Optimization	Katib primary; Ray Tune alternative	Controlled hyperparameter search	Standalone Katib sample
Training	PyTorch, Hugging Face, PEFT LoRA	Model adaptation and checkpoints	Real tiny PyTorch CPU weights tested; Meta Llama PEFT GPU unverified
Observability	JSONL, optional MLflow	Metric history, experiments, artifacts	JSONL verified
AI intelligence	Meta Muse Spark advisor	Bounded numeric recommendations and deterministic fallback	Mock end-to-end + mocked transport tests; live API unverified
Trust boundary	Python deterministic policy	Allowlist, ranges, budgets, approval	Patch validation working
Quality	Holdout evaluation + model registry	Promotion assessment	Future phase

Component ownership

Kubeflow orchestrates; Katib optimizes; MLflow records; Meta advises; the policy engine authorizes. Ray Tune may replace Katib for selected experiments, but two independent HPO systems must never own one trial simultaneously. The agent must not read user secrets or raw training records by default.

2. Recovery and stopping lifecycle



Autonomous bounded training and recovery state machine

Controller states

State	Interpretation	Expected action
Insufficient evidence	Early run or missing history	Continue collecting reliable evaluations
Healthy / progressing	No stopping trigger	Continue until planned completion or budget
Suspected overfit	Validation worsens while train loss improves	Preserve best checkpoint; consider more regularization
Suspected underfit	Both losses plateau high relative to task-informed baselines	Diagnose data, capacity, LR and training budget
Diverged	NaN/Inf or unstable optimization	Immediate safe stop; do not blindly resume
Failed	Exception, missing artifact, OOM, policy violation	Audit and escalate or apply a specifically allowed retry
Completed	Attempt finished without stopping trigger	Validate model independently before promotion

Phase 3 now implements a local durable CPU-mock state journal with idempotent receipt reconciliation, explicit blocked states and an operator-run recovery procedure. This is not distributed Kubernetes job cancellation.

Why the best checkpoint matters

Validation loss may be lowest *before* overfitting becomes evident. Save the checkpoint corresponding to the minimum validation loss, rather than just the last epoch. If a hyperparameter change invalidates the optimizer state or adapter structure, begin a new attempt from the base model instead of assuming that warm-starting is safe. A synthetic JSON marker is **not** a restorable weight checkpoint.

3. Training metrics and diagnostic concepts

Training loss measures fit to training examples; **validation loss** helps assess generalization on held-aside validation examples. Their difference is a heuristic signal, not a universal definition of overfitting.

Illustration: overfitting

Train loss trends downward while validation loss first improves and then rises. It can be reasonable to stop only after sustained validation deterioration and a meaningful train/validation gap. Small or noisy datasets can make such heuristics unreliable.

Illustration: underfitting

Both train and validation loss can remain high or flat. Possible causes include too few updates, excessive regularization, weak representation capacity, mismatched labels, bad data, scheduler problems or an insufficient learning rate. A single numerical threshold cannot reliably diagnose underfitting across tasks.

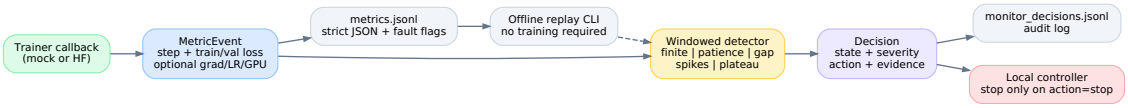
Illustration: divergence

Non-finite loss, gradient overflow or unstable updates merit a safe stop and investigation of learning rate, scaling, data corruption and numeric precision.

Signal	Computation or source	Typical use
Train loss	Trainer callbacks	Optimization progress
Validation loss	Independent validation batches	Checkpoint selection and HPO objective
Generalization gap	`val_loss - train_loss`	Possible overfitting evidence
Validation perplexity	`exp(val_loss)` for token-level cross entropy	Language modeling context only
Accuracy, macro-F1, AUROC	Held-out evaluation tailored to task	Task quality; cannot be inferred from loss alone
LR and gradient norm	Optimizer and trainer hooks	Optimization stability
GPU memory/utilization	Driver and telemetry	Efficiency / OOM prevention
Tokens/sec, GPU hours, spend	Training logs and billing telemetry	Throughput and budget ceilings

Phase 2 now provides: a typed per-evaluation metric contract, warning versus stopping actions, explicit evidence, limited conservative spike/plateau rules, JSONL auditing and offline replay. True step-aligned training windows, generic task metrics and independent GPU probes remain future work.

3A. Phase 2: telemetry pipeline and operational diagnosis



Phase 2 evaluation telemetry, audit logs, decision policy and replay flow

Implementation inventory

Code path	Purpose	Operational status
`autotune/metrics.py` - `MetricEvent`	Typed evaluation sample: positive step, train loss and validation loss; optional timestamp, LR, gradient and GPU peak memory	Tested with synthetic and invalid inputs
`autotune/metrics.py` - `diagnose`	Windowed conservative diagnosis, severity/action classification and numerical evidence	Tested
`autotune/metrics.py` - `MonitorStream`	Accumulate observations, emit a decision for every evaluation to JSONL	Integrated and tested in mock trainer
`autotune/tracking.py`	Strict JSON-safe metric history, nonfinite markers with null values, optional MLflow	JSONL tested; external MLflow not tested
`autotune/monitor_cli.py`	Replay `metrics.jsonl` offline into a comprehensive JSON decision report	Tested
`autotune/mock_trainer.py`	Feed live mock events through monitor and stop only on `action=stop`	End-to-end simulation verified

Code path	Purpose	Operational status
`autotune/hf_trainer.py`	Same monitor call during HF evaluation callback	Source adapted; no real GPU test

Event and decision contracts

```
{
  "step": 6,
  "train_loss": 0.98,
  "val_loss": 1.79,
  "learning_rate": 0.0001,
  "grad_norm": 12.2
}
```

Each event represents the last available train metric paired with an evaluation metric. Required: positive increasing `step`, numeric `train_loss`, numeric `val_loss`. Optional fields: timestamp, learning rate, gradient norm, peak GPU allocation. The logger provides an external UTC audit timestamp. Future work should align train-loss averaging windows precisely to evaluation steps in the HF adapter; the current `on_log` snapshot can lag the validation window.

Example decision structure, stored in `monitor_decisions.jsonl`:

```
{
  "state": "overfit",
  "explanation": "Validation loss rose for 2 evaluations while train loss fell; gap=0.810",
  "severity": "critical",
  "action": "stop",
  "evidence": {"step": 6, "gap": 0.81, "patience": 2, "best_val_loss": 1.57}
}
```

Stop semantics are explicit: the training callback must inspect `action`, not assume every named anomaly means a stop. `info` and `warning` decisions continue training. Severe errors such as nonfinite loss can request termination immediately even before a warmup window completes.

Diagnostic matrix and false-positive protection

Signal	Necessary evidence	Classification	Current behavior
Warmup	Fewer than `min_evaluations` paired events	Info / continue	No stop
Missing or repeated steps	Missing metric or non-monotonic evaluation sequence	Warning / continue	Signals insufficient evidence
Nonfinite value	NaN/Inf loss or gradient norm	Critical / stop	Trial stop; no automatic divergence patch yet

Signal	Necessary evidence	Classification	Current behavior
Isolated train+val spike	Both losses jump above configured ratio in one evaluation	Warning / continue	Require confirmation
Repeated train+val spikes	Multiple consecutive jumps exceed `loss_spike_ratio`	Critical / stop	Stop to avoid uncontrolled divergence
Overfit heuristic	Validation rises every patience observation; train improves; gap exceeds threshold	Critical / stop	Preserve best simulated checkpoint marker
High flat losses	Both losses remain high and near-flat for full window	Critical / stop	Task-specific floor may need calibration
Low flat losses	Both losses flat but not over underfit floor	Warning / continue	Not proof of underfitting
High finite gradient norm	Latest grad norm exceeds `grad_norm_warning`	Warning / continue	Not auto-stop without trend evidence

Underfit and overfit labels are *heuristic findings*; they do not establish that the model is truly underfitting or overfitting across all tasks. A genuine research or production project should calibrate thresholds on data scale, dataset size, model family and training objectives.

Offline replay / reproducibility

From VS Code PowerShell:

```
.\.venv\Scripts\python.exe -m pytest -q
.\.venv\Scripts\python.exe -m autotune.orchestrator --mode mock --no-writeback
.\.venv\Scripts\python.exe -m autotune.monitor_cli --file "runs\YOUR_RUN_FOLDER\trial-01\metrics.jsonl"
        --config config/baseline.yaml --output "runs\monitor_report.json"
```

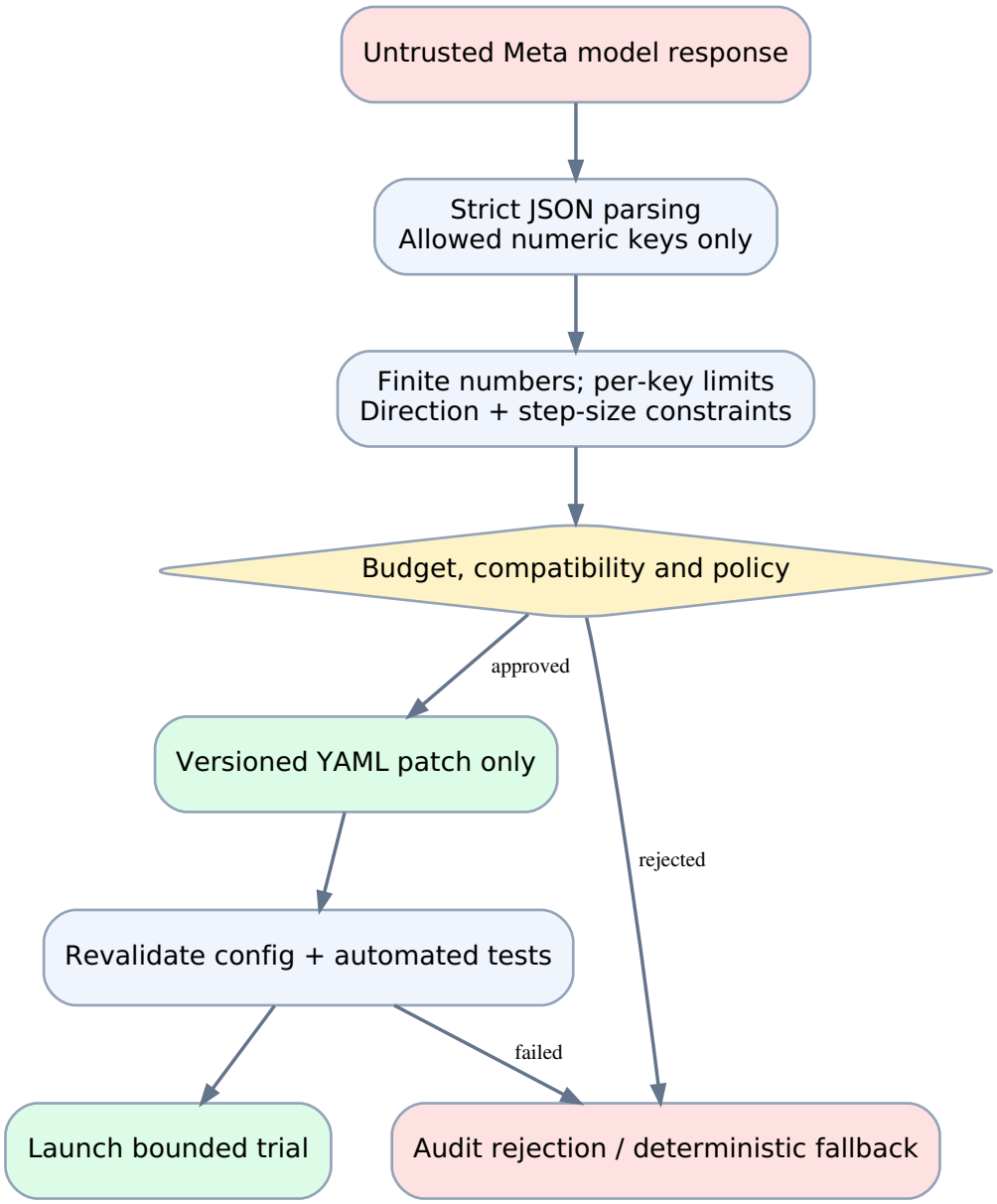
Expected benchmark-free demonstration: 36 local Python tests passed; synthetic overfit at step 6 in attempt one, a second attempt completes, with best **simulated** val loss 0.58. Actual Llama GPU fine-tuning and Kubeflow/MLflow production connections are not evidenced here.

Known technical limitations and next phase

1. `MetricEvent` validates only essential loss fields and limited optional signals; no universal accuracy, F1 or dataset-quality metrics yet.
2. Date/timestamp is recorded but ISO timestamp parsing / clock drift validation is not enforced.
3. Very noisy datasets can defeat monotonic-patience rules; tune thresholds with task-specific baselines before GPU usage.

4. Missing metric records do not stop, though a production controller should escalate after a configurable missing-data timeout.
5. The policy fallback does not auto-remediate divergence. The Phase 3 durable mock controller fails closed instead of retrying blindly.
6. `monitor_decisions.jsonl` is per-trial audit evidence; Phase 3 now adds atomic LOCAL journal and simulated checkpoint validation. Distributed job reconciliation and real checkpoint validation remain future work.

4. LLM safety and codebase change boundary



Trust boundary between model suggestions and executable training decisions

The advisory agent returns a structured object such as:

```
{
  "patch": {
    "learning_rate": 0.00008,
    "lora_dropout": 0.15,
    "weight_decay": 0.03
  }
}
```

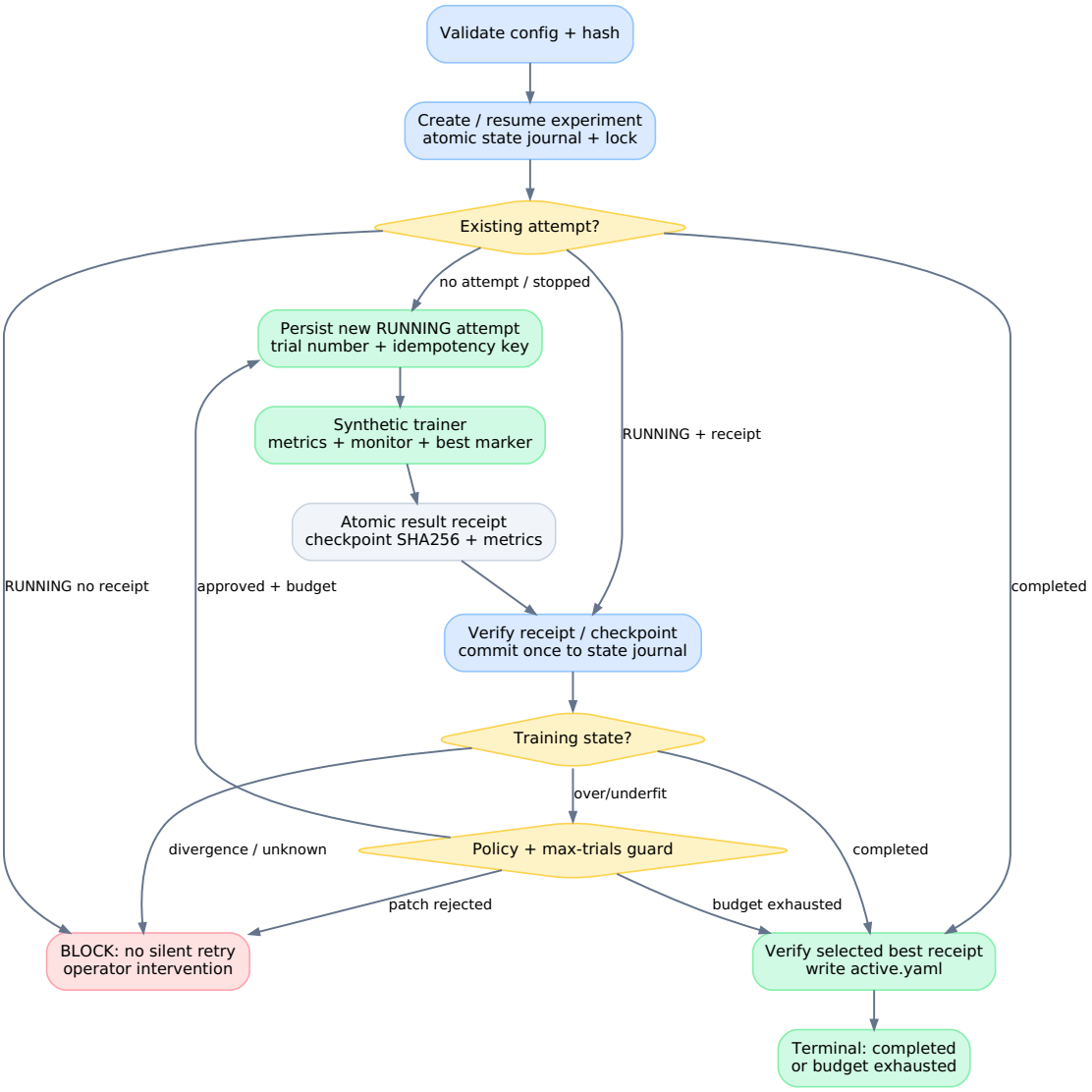
An agent proposal is not permission. The engine checks allowed keys, finite values, absolute bounds, maximum per-attempt change and diagnosis-specific directional rules. Existing v0.1/v0.3 policy supports `learning_rate`, `weight_decay`, `lora_dropout`, `num_train_epochs`, `lora_r`, and `gradient_accumulation_steps`. Any unknown parameter or executable instruction is rejected.

Phase 9 will add **Git branch -> patch -> CI -> review/approval -> merge -> restart** for configuration or limited code changes. We must not let an LLM arbitrarily rewrite `.py`, import untrusted training scripts or run commands. When remote Meta access fails, the deterministic fallback can be used if policy accepts it.

Operational stop conditions

- Retry ceiling reached; no approved patch; no trustworthy checkpoint; invalid/nonfinite metrics.
- GPU or financial budget exceeded; unsupported device precision; repeated OOM or external API failure.
- Unapproved model, unlicensed dataset, data leak risk or failed quality/evaluation gate.

4A. Phase 3: durable and safe recovery (v0.4)



Local journal, checkpoint receipt, idempotent reconciliation and blocked-state control

Why persistence is necessary

In Phase 2 an in-memory process controlled several simulated trials. A process kill could happen after training finished but before an attempt was recorded, resulting in duplicate GPU spending or invalid artifact selection. In Phase 3 each attempted operation has a durable state before starting and an atomic result receipt after finishing. Resume reconciles these two independent records rather than blindly repeating work.

State and transition contract

State or marker	Written when	Resume behavior
`running` attempt	Before invoking the trainer	Inspect `result_receipt.json` first
Valid result receipt	After the trainer returns, with checkpoint SHA256	Commit exactly once, never rerun finished attempt
`stopped`	Verified monitor signal indicates over/underfit or another stop	Only over/underfit with approved patch can retry
`completed` attempt	Finished training without monitored stop	Experiment terminates and checks best checkpoint
`abandoned`	Operator confirmed an unreceipted prior job ended	Trial number counts against budget; retry is explicit
`blocked` experiment	Uncertain job, bad receipt, invalid checkpoint or unsafe condition	Human investigation; no automatic retry
`budget_exhausted` experiment	Trial count reaches max while verified candidate exists	Keep best candidate but no release promotion
`completed` experiment	A complete attempt exists and best marker is verified	No-op on repeated resume

Idempotency: each attempt has a stable SHA256 idempotency key combining experiment ID, trial number and configuration hash. The journal at `recovery_state.json` is the authoritative record; `recovery_events.jsonl` supports incident analysis but can lag a journal transition if the host crashes at the wrong moment. Completion receipts include attempt identity, config SHA256, simulated checkpoint SHA256 and best validation metrics. A correct receipt may be safely committed after restart.

Crash uncertainty: with state=`running` and no valid receipt, the recovery engine will refuse to start another attempt automatically. The operator must verify no training job is alive, then use an explicit abandon action; it consumes a trial slot. Untrusted or mismatched receipts fail closed. A local exclusive `.controller.lock` blocks concurrent writers; clearing a stale lock requires explicit operator acknowledgement that no controller remains active.

Source and executable interfaces

Module / file	Responsibility
`autotune/recovery.py`	Local state machine, atomic JSON, locks, receipts, checksum verification, retry and best-state selection
`autotune/recovery_cli.py`	Windows-friendly new-run and resume CLI; operator interventions require flags
`tests/test_recovery_phase3.py`	Fault injection, idempotent completion, corruption, lock refusal, baseline-change refusal, budget and opt-in writeback
`docs/PHASE3_RUNBOOK.md`	Operator recovery procedure, command examples and safety boundaries
`run_phase3.ps1`	Doctor, test suite and durable synthetic demo

Implementation sequence and restart behavior

1. Validate baseline YAML and calculate its stable SHA256 before creating a fresh experiment.
2. Write a journal, atomically write each attempt config, acquire the experiment lock and record the `running` attempt before invoking the trainer.
3. Receive evaluation metrics from the existing Phase 2 monitor and let the mock trainer create its best-checkpoint **JSON metadata marker**.
4. Hash the artifact and write an atomic `result_receipt.json`; match the trial ID, config hash, best step and loss against checkpoint metadata.
5. Atomically commit the verified result and update `best_config.yaml` only when validation loss improves; if stopped for overfit/underfit, apply a bounded deterministic patch and start another numbered attempt.
6. On completion or exhausted budget, re-verify the winning receipt/marker before writing `active.yaml`. Never run test-set evaluation or automatic model deployment at this point.
7. If interrupted, reconcile a valid receipt without rerunning; block if work is uncertain, corrupted or unsupported. Resume requires the same baseline config fingerprint.

```
# Phase 3 new CPU-mock experiment
.\.venv\Scripts\python.exe -m autotune.recovery_cli --config config/baseline.yaml --workspace runs

# Reconcile a previous experiment after verifying no active external job
.\.venv\Scripts\python.exe -m autotune.recovery_cli --config config/baseline.yaml --resume
        'runs\YOUR_EXPERIMENT'

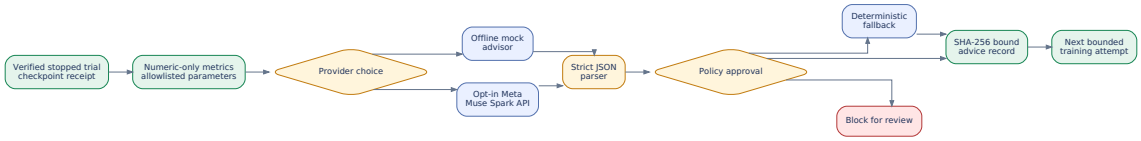
# Only if prior job definitely stopped and no receipt exists
.\.venv\Scripts\python.exe -m autotune.recovery_cli --config config/baseline.yaml --resume
        'runs\YOUR_EXPERIMENT' --abandon-interrupted
```

Validated tests and limitations

Verification: 48/48 tests passed locally on Python 3.13, adding 12 Phase 3 tests to the 36-test v0.3 suite. Simulated failure points include after recording RUNNING, after training but before receipt, and after receipt but before journal commit. Valid receipts reconcile without duplicated metrics. Corrupted checkpoint hashes prevent selection; the abandoned trial counts against the retry ceiling. A second identical resume on a terminal experiment does not emit new events.

Limits: Phase 3 only supports synthetic CPU training through the new durable CLI. It does not yet implement durable PyTorch weight checkpoints, cross-process Kubernetes job reconciliation, distributed locks, real hardware cost quotas, codebase Git commits or live Meta API reasoning. `SHA256` proves local bytes match a recorded digest, not that the model is trustworthy or loadable. The old `autotune.orchestrator` is retained for backward compatibility, but it is not the new crash-safe controller. Testing on the user's Windows host is still required.

4B. Phase 4: Meta advisor integrated into durable recovery (v0.5)



Strict advisory trust boundary and retry integration

What was actually implemented

Phase 4 introduces `autotune/advisor.py` and `autotune/advisor_cli.py`, connects validated advice to Phase 3's `run_durable` controller, and adds `config/agent_mock.yaml` for zero-cost CPU validation. The same recovery loop can be configured for a hosted Meta Muse Spark recommendation, but **a successful real external API call has not been executed or claimed**. The default `config/baseline.yaml` continues using the pre-existing deterministic rule engine, protecting Phase 3 compatibility.

The Meta endpoint and model identifiers are drawn from Meta's public Model API documentation (checked 2026-11-01): `https://api.meta.ai/v1` and `muse-spark-1.3`, through an OpenAI-compatible Chat Completions request. An API key remains in the `MODEL_API_KEY` environment variable, never in the training configuration. The optional network call is bounded by a 20-second client timeout and zero client retries. Offline mock and network failure do not require credentials.

Interface contracts

Contract	Input	Valid output	Rejection / fallback
Advisor input	Diagnosis + last eight allowlisted metric dictionaries + six numeric training parameters	Sanitized JSON only	Raw samples, commands, file paths and secrets never included
Model response	Strict JSON object with `patch`, optional matching `diagnosis` and short `rationale`	Numeric patch with allowlisted keys	Non-JSON, duplicate JSON keys, unknown fields, oversized output, contradictory diagnosis and bad types rejected
Policy engine	Proposed patch + current configuration + original monitor diagnosis	Bounded direction-aware hyperparameter delta	Unapproved values and changes rejected, deterministic safe fallback attempted
Durable snapshot	Trial number, config SHA-256, checkpoint SHA-256, validated patch, output config SHA-256	`trial-XX/advisor_decision.json` with its own hash	Altered or inconsistent saved decision blocks replay
Controller	Validated candidate configuration, finite attempt budget	New versioned CPU-mock training attempt	Divergence or unsupported cause blocks automation

Why this boundary matters

A language model can output malformed JSON, non-finite numbers, duplicate keys, off-policy parameter changes, or text designed to trigger dangerous actions. Phase 4 handles all model replies as **untrusted data**: the assistant cannot request terminal access, execute arbitrary code, see training records or secrets, edit Python files, or authorize promotion. A separate Python validator checks every suggested parameter. The saved recommendation is bound to the exact stopped attempt and checkpoint and is **replayed rather than recomputed** after an interruption.

The deterministic fallback does not promise better performance: it only makes a bounded, policy-consistent candidate. The model must still be trained and evaluated before any quality conclusion. A CPU mock scenario uses synthetic losses and a JSON metadata checkpoint, not real gradients or Meta Llama weights.

Example synthetic recovery trace

1. Trial 1 emits falling train loss but rising validation loss after its optimum; the monitor stops it with an overfit diagnosis.
2. Phase 3 validates the mock checkpoint marker digest and commits the receipt to durable state.

3. The Phase 4 mock advisor proposes `lora_dropout: 0.15`, `weight_decay: 0.03`, and `learning_rate: 0.00008` from starting values `0.05`, `0.01` and `0.00010`.
4. The deterministic policy verifies values and directions. The controller writes `trial-01/advisor_decision.json` with both config hashes and the checkpoint hash.
5. The controller starts trial 2 from the baseline simulator with the new configuration. The scripted loss curve ends at validation loss **0.58**, which is **synthetic** and not a real model accuracy claim.
6. On restart, a saved advisor decision is independently revalidated; a tampered snapshot blocks instead of making a fresh API call and overwriting audit history.

CLI implementation and reproducibility

```
py -3.13 -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r requirements.txt
.\.venv\Scripts\python.exe -m pytest -q
.\.venv\Scripts\python.exe -m autotune.advisor_cli --provider mock --diagnosis overfit
.\.venv\Scripts\python.exe -m autotune.recovery_cli --config config/agent_mock.yaml --workspace runs
```

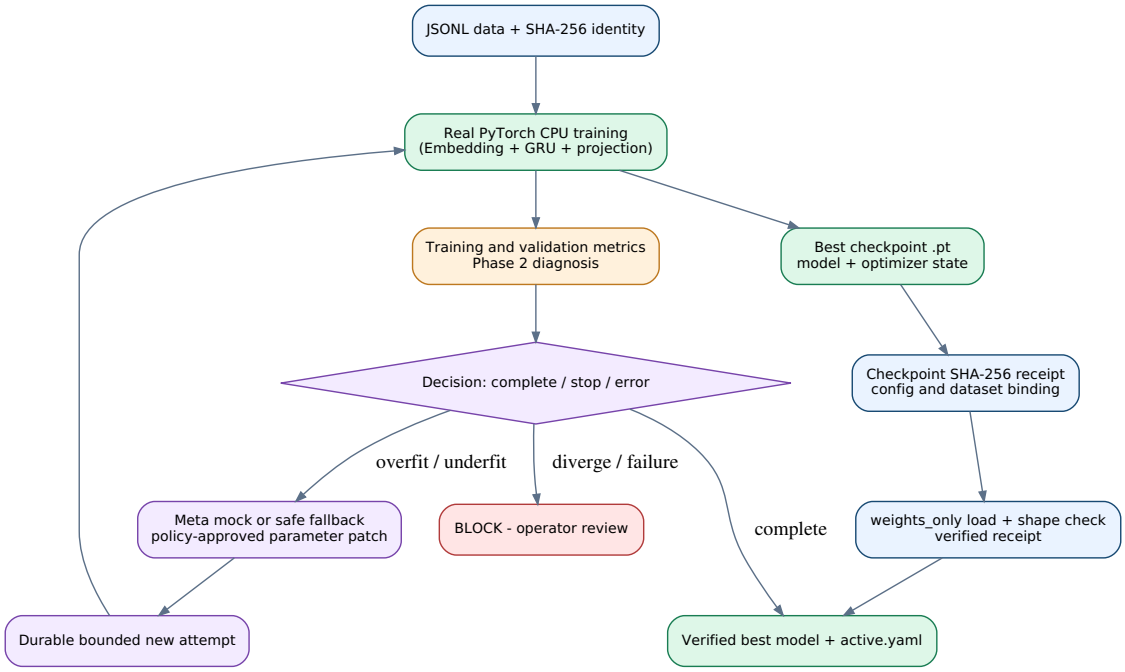
An optional hosted test is **explicit**: set `MODEL_API_KEY` securely and invoke `python -m autotune.advisor_cli --provider meta --diagnosis overfit` (after installing the optional `openai` client). API access might be billable; absence of a key produces a visible deterministic fallback, not fabricated remote inference. See `docs/PHASE4_RUNBOOK.md` for Windows and PowerShell specifics.

Verification evidence: 77/77 local Python tests passed, including 29 new Phase 4 tests for malformed/adversarial responses, injection-safe payload selection, missing credentials, replay consistency, tamper rejection and an end-to-end synthetic durable recovery. The actual hosted endpoint was not contacted. No GPU, Kubeflow cluster, actual weight checkpoints, or real measured improvements were used.

4C. Phase 5A - Genuine PyTorch training with verified weight checkpoints (v0.6)

> **Verification scope:** A small byte-level GRU language model was genuinely trained on the bundled JSONL instructions using offline PyTorch CPU, with real backpropagation and optimizer updates. The optional Hugging Face Meta Llama LoRA adapter in `autotune/`

> `hf_trainer.py` is still NOT GPU-tested; this is not a claim of real Meta Llama fine-tuning or state-of-the-art quality.



Real weight training, auditing, and bounded recovery

Why this phase is different from the earlier simulation

Phases 1-4 replayed deterministic simulated metrics and stored `best_checkpoint.json` as **metadata**, not model weights. Phase 5A introduces `autotune/real_training.py`, which computes token-level cross-entropy, calls `loss.backward()`, updates parameters with AdamW, evaluates on a separate validation set and saves actual tensors to `best_checkpoint.pt`. The byte model uses an embedding, a GRU and a linear vocabulary projection; it is randomly initialized and is not a pretrained Meta model. Phase 5A does **not** claim fine-tuning of pretrained weights.

End-to-end implementation

1. **Dataset access and provenance.** Read JSONL records with a mandatory `text` field. Require at least two records per split, separate training and validation content, and a bounded local file size. Calculate a SHA-256 identity over both file digests and their explicit split roles. Do not send text records to the advisor.
2. **Tokenization and labels.** Encode UTF-8 into byte IDs (0-255). Input tokens predict their next byte. Mask padding positions with

`-100` in cross-entropy. This is an intentionally lightweight architecture for testing rather than a production text tokenizer.

3. **Weight updates.** Train with AdamW, gradient clipping and deterministic example order. Each epoch produces an average training loss and a separately computed validation loss. The training loss is averaged during the epoch, not recomputed after the epoch; it is a trend estimate rather than perfectly synchronized with validation loss.
4. **Signal monitoring.** Write metrics to `metrics.jsonl`, diagnostic decisions to `monitor_decisions.jsonl`, and run the existing evidence-window rules to stop at an epoch boundary when warranted. These heuristics may misclassify short or noisy runs.
5. **Checkpoint save.** On each improvement, atomically replace `best_checkpoint.pt` with model weights, optimizer state, epoch, validation score, configuration hash and dataset hash. This supports verification and loading; partial-epoch optimizer resume is not yet implemented.
6. **Durable controller.** `autotune/real_recovery.py` uses a separate `real-torch-v0.6` journal schema, creates a receipt containing the exact file SHA-256 and selects a best checkpoint only after verifying checksum, identities, state dictionary shapes and optimizer loadability with `torch.load(weights_only=True)`.
7. **Controlled recovery.** A stopped overfit/underfit attempt can request an allowlisted patch through the existing offline advisor/fallback engine and start a new bounded attempt. The model restarts from a fresh initialization when parameters change. Uncertain RUNNING work without a receipt is blocked; no silent re-execution occurs. Divergence also blocks rather than blindly retrying.

Actual measured CPU smoke test

Epoch	Train loss	Validation loss
1	5.5887	5.6072
2	5.5706	5.5941
3	5.5538	5.5808
4	5.5384	5.5675

The smoke test completed one trial and selected epoch 4; its actual tensor checkpoint was hash-verified and loadable. These values demonstrate execution only, not model quality or comparison with an established baseline. A deliberate threshold-manipulation test also

exercises two real-weight attempts and budget exhaustion; its *underfitting classification is forced for state-machine coverage*, not proof of a genuine learning failure.

Verification evidence: 95/95 automated tests passed locally with PyTorch CPU (77 inherited plus 18 new tests), including real gradient backpropagation, JSONL/dataset integrity, weight and optimizer checkpoint loading, digest/tamper failures, completed-run idempotency, dataset mutation rejection, uncertain RUNNING blocks and bounded retry. The user subsequently confirmed Phase 5A on Windows: 95 tests passed and a loadable tensor checkpoint was produced.

Run on Windows in VS Code

```
py -3.13 -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r requirements-real-cpu.txt
.\.venv\Scripts\python.exe -m pytest -q
.\.venv\Scripts\python.exe -m autotune.real_cli --config config/real_cpu.yaml --workspace runs_real
```

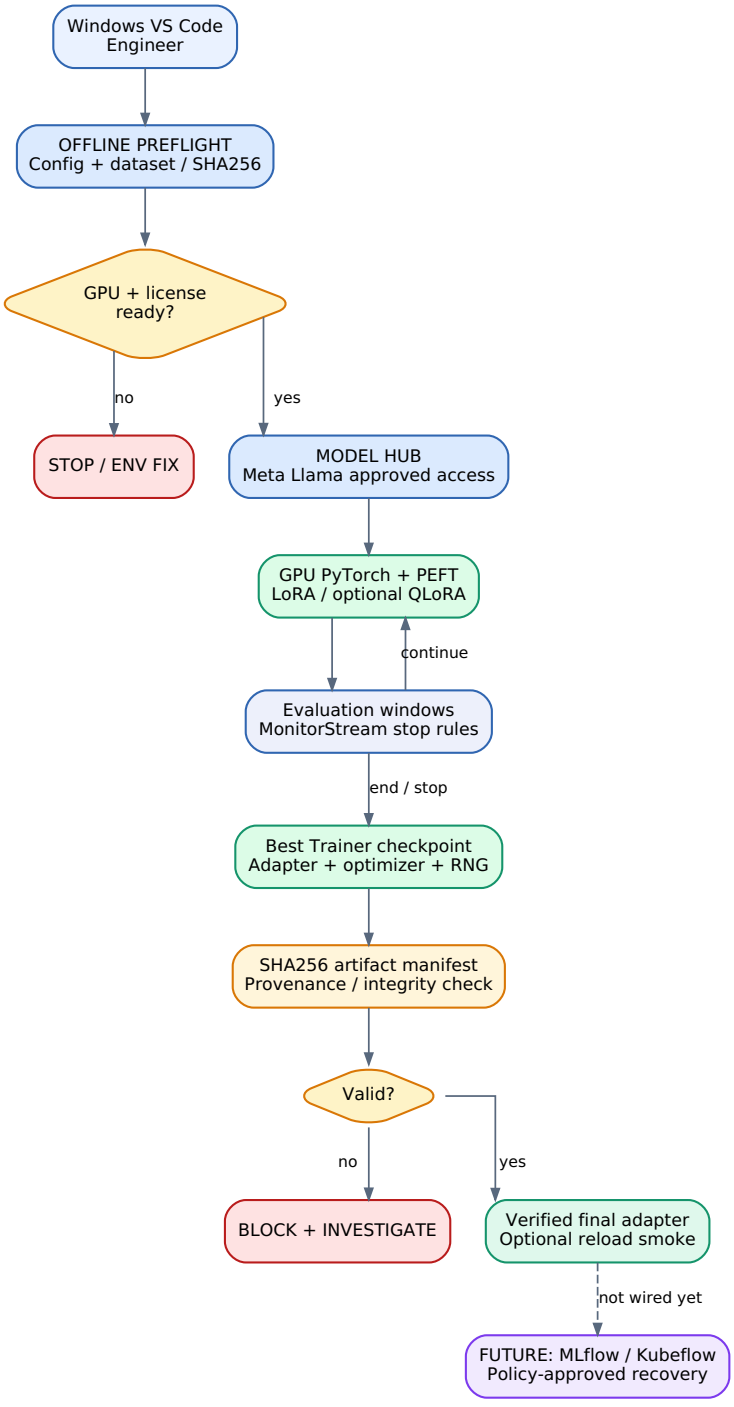
If script execution is allowed, `._phase5.ps1` runs the tests and CPU training. Do not set system-wide unrestricted PowerShell execution policies merely to run it. Inspect `runs_real/<id>/trial-01/best_checkpoint.pt`, `result_receipt.json`, `recovery_state.json`, `active.yaml`, and event/metric logs. The phase requires PyTorch and may need a compatible wheel for Windows/Python; no API key, GPU or network model download is necessary after installation.

Stage 5B background and outstanding live Meta Llama validation gate

The existing `autotune/hf_trainer.py` is **source only and not GPU validated**. Before claiming full Phase 5, exercise a licensed Meta Llama variant with the supported Transformers, PEFT, Accelerate and GPU stack, validate tokenizer and masking semantics, use aligned metrics, verify adapter reload/inference, save optimizer/scheduler state where supported, handle interrupted trials and record GPU memory and data/model revision hashes. The CPU GRU checkpoint is **not interchangeable** with HF/LoRA checkpoints. Neither pipeline is production release-ready.

4D. Phase 5B - Meta Llama / Hugging Face LoRA integration (v0.7)

> **Verification boundary:** The new LoRA pipeline is source-complete for an optional, explicitly triggered pilot and passed offline contract tests. No Meta model weights were downloaded, no GPU fine-tuning was run, and Hugging Face Trainer’s real checkpoint behavior has not been exercised in this build. The user’s Phase 5A Windows run (95 tests) trained the separate tiny-GRU model, not Llama. This release advances integration readiness, not a completed on-GPU milestone.



Guarded Meta Llama LoRA pathway and future MLflow/Kubeflow bridge

Why LoRA and QLoRA are useful

LoRA (Low-Rank Adaptation) freezes a pretrained base model and trains a small number of low-rank update matrices in selected layers. This reduces the number of trainable parameters compared with full fine-tuning. QLoRA loads base weights at reduced precision (commonly

4-bit) while learning LoRA adapters. Neither approach guarantees lower loss or better model quality; memory efficiency and stability depend on the actual hardware, libraries and dataset.

On a GPU server, the plan selects a Meta Llama model, validates authorized model access, tokenizes train/validation JSONL, attaches PEFT adapters to `q_proj` and `v_proj`, and runs Hugging Face Trainer. The offline plan does not make external requests and rejects missing/duplicate splits and configurations that schedule no validation. A separate public `sshleifer/tiny-gpt2` CPU config provides an optional online dependency smoke test.

New code and ownership

Module	Responsibility	Verified scope
`autotune/lora_contract.py`	Offline config/data hash check, split duplication detection, evaluation schedule plan, SHA256 checkpoint manifests	39 new offline tests together with CLI/guardrails
`autotune/lora_cli.py`	Default safe preflight; explicit `--execute`, `--resume-checkpoint`, `--verify-adapter`	Offline planning tested
`autotune/lora_gpu.py`	PEFT LoRA/optional QLoRA, GPU guard, HF Trainer callbacks, adapter save/reload	Source-level integration; real HF run not yet tested
`config/lora_meta_gpu.yaml`	Licensed Meta Llama 3.2 1B pilot config	Validated, not downloaded
`config/lora_public_cpu.yaml`	Public tiny GPT-2-derived CPU training example	Validated, not downloaded
`docs/PHASE5B_RUNBOOK.md`	Windows and optional remote AMD ROCm/NVIDIA GPU steps	Documented, not executed

Correct metrics and checkpoint semantics

The Hugging Face callback collects a training-loss log at the same evaluation cadence (currently `logging_steps == eval_steps`). At evaluation, it pairs the mean of newly logged training-loss values with the validation loss. If no corresponding training loss exists, it writes an `unpaired_evaluation` event and **does not invent a pair or issue a stop**. The monitor uses existing conservative heuristics and can request the Trainer to stop and save at the next safe control boundary.

A *full training checkpoint* includes an adapter `adapter_model.safetensors`, `adapter_config.json`, `trainer_state.json`, `optimizer.pt`, `scheduler.pt`, and RNG state. The separate `final_adapter/` is suitable for inference reload but **is not guaranteed to restore training optimizer state**. Hash manifests bind artifact files to

configuration/model/dataset identities and reject missing, extra or modified files. Hashes are not signatures; untrusted optimizer pickle files must never be resumed.

Important integration boundary: The old Phase 5A tiny-GRU `.pt` checkpoint and the new HF Trainer/PEFT checkpoint are not interchangeable. The HF pilot is not yet attached to Phase 3/4 durable self-healing retry logic. An operator must explicitly start a GPU run and explicitly select a trusted in-run checkpoint to resume. Automatic GPU parameter tuning and requeue are future work.

Agent boundary, model licensing and compute

Meta Muse Spark remains a *diagnostic advisor* in the existing CPU controller, never a shell executor or direct repository editor. It is **not automatically called** by the new HF pilot. Meta Llama model downloads require approved license access through Hugging Face. The base model revision is configurable; default `main` is a moving reference and must be pinned to a commit for strict reproducibility. Trusting remote model code is disabled in the pilot.

For AMD MI300X, select compatible Linux ROCm PyTorch wheels and validate the relevant PEFT and bitsandbytes build before enabling 4-bit QLoRA. `torch.cuda.is_available()` is also used by ROCm-enabled PyTorch; a Windows CPU installation alone cannot satisfy GPU deployment. For NVIDIA use a compatible CUDA stack. No GPU throughput, GPU memory or cost results are claimed in this edition.

Phase 5B test record and Windows validation

The user independently verified **95 passed in 34.36s** on their Windows host in Phase 5A, plus a real tiny-GRU best validation loss `5.567494869232178` and SHA256 checkpoint receipt. A PyTorch NumPy warning was nonfatal, so the new `requirements-real-cpu.txt` includes `numpy`. This v0.7 build adds 39 offline tests, bringing the build environment to **134/134 passing**. Tests cover invalid LoRA configurations, missing validation windows, data leakage, tampered/missing artifact files, hash-bound provenance, symlinks, locking, GPU absence and token label padding. **Fake test artifact bytes do not constitute trained adapter weights.**

On Windows in VS Code (no HF model download):

```
.\.venv\Scripts\python.exe -m pip install numpy
.\.venv\Scripts\python.exe -m pytest -q
.\.venv\Scripts\python.exe -m autotune.lora_cli --config config/lora_meta_gpu.yaml
```

Optional public online CPU smoke and GPU execution commands are detailed in `docs/PHASE5B_RUNBOOK.md`. Execute those before marking the actual GPU/adapter gate complete. This is **not a production benchmark or safety certification**.

Risks and next release gate

The live HF/PEFT dependency matrix, Meta repository permissions, compatibility of checkpoint file layouts across Trainer versions, ROCm/CUDA kernels, actual optimizer resume, model inference, GPU OOM and VRAM metrics remain to be validated. The training callback is conservative and does not prove that two correlated loss numbers represent equivalent distributions. Final quality evaluation must use an untouched holdout and a separately approved promotion policy.

Next milestone: Phase 6 after a verified LoRA GPU run - connect real adapter artifacts and evaluation metrics to a persistent MLflow tracking server; only then integrate those jobs with durable Kubeflow-owned recovery. Continue to refresh the PDF handbook, roadmap and AGENTS.md for every active iteration.

5. Full phased implementation program

Phase	Scope / deliverable	Exit gate	State
0	Requirements, architecture, docs	Baseline agreed, living docs present	Complete
1	Package, validation, doctor, tests	Local mock and tests pass	Complete
2	Metric contract, diagnostics	Noisy/missing/divergent cases tested	Complete - 36 total tests at Phase 2
3	Recovery state machine	Idempotent stop/ checkpoint/retry tests	**Complete - 48 total tests**
4	Meta diagnostic agent	Strict JSON and adversarial-output tests	Complete - 77 total CPU-mock tests at Phase 4
5	Real PyTorch CPU (5A) + Llama LoRA (5B)	Loadable CPU weights; then supported GPU adapter validation	**5A user-verified; 5B contract tested (134 total); live GPU pending**
6	MLflow infrastructure	End-to-end run lineage and artifact store	Planned
7	Kubeflow deployments	Cluster pipeline launches bounded attempts	Planned
8	Katib / Ray Tune	Verified optimizer selection and trial search	Planned

Phase	Scope / deliverable	Exit gate	State
9	Git-reviewed changes	PR, tests, rollback, approval policy	Planned
10	Unified dashboard	Live metrics, timeline, controls and alerts	Planned
11	Quality/policy gates	Held-out evaluation and model registry	Planned
12	Production hardening	Reproducible installation and tested runbook	Planned

Critical path: Phases 0-4 completed as CPU mock; Phase 5A real tiny PyTorch weights verified. **Next:** Phase 5B live GPU verification, then Phase 6 MLflow. Add MLflow persistence before moving recovery onto independent Kubernetes jobs; then add Katib. The dashboard can grow incrementally but the final promotion gate follows robust evaluation.

Deliverable acceptance by phase

Phase 2: typed metric events, paired train/eval event contract, robust anomaly tests, diagnosis evidence.

Phase 3 (mock complete): durable local transitions, attempt keys, simulated checkpoint SHA256 verification, failure injection, safe resume. Real weight-checkpoint compatibility remains Phase 5+ work.

Phase 4 (CPU mock complete): strict Meta response schema, offline integration, privacy allowlists, failed-call fallback, persisted advisor decision and tamper/replay tests; hosted API connection still unverified.

Phase 5A: real small PyTorch CPU weights + checkpoint validation verified. **Stage 5B outstanding:** actual Meta Llama LoRA adapter weights and repeatable validation loss from a supported GPU.

Phases 6-8: MLflow tracking, Kubeflow job isolation and Katib-controlled search with safe ownership.

Phases 9-12: governed Git changes, UI, holdout and model registry, deployment/performance/security checks.

6. Phase 1 - implementation record

Starting point: Existing v0.1 CPU-only synthetic controller, metric heuristics, patch allowlist, optional Meta/MLflow hooks, HF LoRA adapter, Kubeflow pipeline definition and Katib example. The user-provided starter's historical tests were 8 passing; no real GPU/KFP deployment had been verified.

Added in v0.3:

1. `autotune/config.py` validates baseline structure and critical parameters before a run starts. It rejects invalid numeric values, booleans masquerading as numeric values, malformed YAML, invalid budgets, invalid LoRA ranks and identical train/validation files. It produces a content-stable SHA-256 config fingerprint.
2. `autotune/doctor.py` exposes `--mode mock|hf` to check the local environment and emit JSON status without printing API secrets.
3. `autotune/orchestrator.py` now loads the validated configuration before creating a run directory and puts the SHA-256 content identity into the experiment manifest.
4. `tests/test_config.py` adds seven tests across contracts, invalid values, no-side-effect failures and environment checks.
5. Master plan, diagrams and maintained AGENTS.md have become first-class project artifacts.

Test results from this session

```
python -m pytest -q
..............     [100%]
15 passed in 0.12s

python -m autotune.doctor --mode mock
"ok": true
```

Synthetic recovery: attempt 1 detected overfitting at evaluation step 6; its best simulated validation loss was 1.57 at step 4. Attempt 2 completed the mock curve with best validation loss 0.58. These numbers are intentionally generated by a deterministic test harness and **must not be cited as real model accuracy or comparative improvements.**

Environment of verification: Python 3.13 in the working container, with installed PyYAML and pytest. Commands for Windows Python 3.11 are provided but have not been executed on the user's machine.

7. How to run Phase 1 in VS Code (Windows)

Extract the project ZIP, open `AutoTune_AI_Phase1` in VS Code and use its integrated PowerShell terminal. From the extracted project root:

```
py -3.11 -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r requirements.txt
.\.venv\Scripts\python.exe -m autotune.doctor --mode mock
.\.venv\Scripts\python.exe -m pytest -q
.\.venv\Scripts\python.exe -m autotune.orchestrator --mode mock --no-writeback
```

The final command creates a new `runs/<timestamp>/` experiment directory with `manifest.json`, `summary.json`, event history, trial metrics, YAML configs and synthetic checkpoint metadata. `--no-writeback` leaves `config/optimized.yaml` unchanged; remove that flag when explicitly testing controlled config writeback.

For optional MLflow + dashboard, install `requirements-optional.txt` and configure a local tracking server as outlined in the README. This is optional for the local simulation; it is not necessary for the 48-test suite.

Common problems and resolutions

Symptom	Likely cause	Next action
`py -3.11` not recognized	Python launcher/Python 3.11 not installed	Install Python 3.11 and verify `py --list`
PowerShell activation blocked	Execution-policy restrictions	Call `.venv\Scripts\python.exe` directly; no activation required
`ConfigError` on baseline	Wrong key/type/bounds	Read error; correct YAML, rerun doctor
Meta advisor unavailable	No API key/network/provider access	Use deterministic fallback; no secret in Git
No evaluation in HF run	Too few training steps vs `eval_steps`	Increase dataset/epochs or reduce eval interval
Missing model checkpoint	Synthetic `best_checkpoint.json` is metadata	Real GPU adapter must be trained before restoration
Kubeflow example cannot run	No compatible Kubernetes/KFP infrastructure	Complete Phase 7 cluster preflight

8. Proposed deployment and GPU strategy

Development: Windows + VS Code, CPU-only simulator, unit tests, local files, Git branches. The first real GPU milestone can run on a supported NVIDIA CUDA environment or a validated AMD ROCm environment, depending on model precision and library compatibility. Never assume a package built for one compute stack works on another without checking drivers, PyTorch and PEFT versions.

Production target: Container registry + Kubernetes namespace, GPU device plugin, Kubeflow Pipelines, Katib, MLflow tracking service and external durable artifact storage. Put dataset/model credentials in Kubernetes Secrets with least privilege. Keep run IDs, dataset revisions, model revisions and auditable policy decisions connected. Introduce GPU-hour budgets, storage limits, cancel paths and approval before auto-deployment.

Data and model governance

Training data must be licensed and separated into train/validation/final holdout partitions. Record dataset fingerprints, task definitions and model revision identities. A final holdout set is evaluated sparingly and cannot feed the agent’s optimization loop. Use model cards and evaluation reports before registry promotion. Phase 5A implements separate train/validation fingerprints; final untouched holdout and production model revision governance remain outstanding.

9. Next task and checkpoint protocol

Next verification gate is running Stage 5B on a supported GPU, then connecting Phase 6 MLflow. Replace synthetic checkpoint markers with loadable model/adapter weights, enforce correct training/evaluation step alignment, save compatible optimizer state when needed, and connect the existing Phase 4 advisor to measured training curves. Live Meta endpoint testing remains an optional separately verified gate.

At the end of every future AutoTune AI project iteration:

1. Run the relevant tests and report true pass/fail evidence.

2. Update `docs/MASTER_EXECUTION_PLAN.md` and the current phase status.
3. Update this handbook source with new diagrams/concepts/implementation and generate the next PDF edition.
4. Append a dated checkpoint, architecture decisions, errors, next steps to `AGENTS.md`.
5. Distribute a new project ZIP and the PDF. Never silently claim an integration was live-tested.

Sources and references

- Kubeflow Pipelines v2 compiling: https://www.kubeflow.org/docs/components/pipelines/user-guides/core-functions/compile-a-pipeline/
- Kubeflow Katib overview: https://www.kubeflow.org/docs/components/katib/overview/
- Katib early stopping design: https://www.kubeflow.org/docs/components/katib/user-guides/early-stopping/
- MLflow experiment tracking: https://mlflow.org/docs/latest/ml/tracking

Remaining Work & Verification Gates

Checkpoint: 01 November 2026 | **Scope:** documentation update only, no changes to source code

Verified: 134/134 offline build tests for v0.7; 95/95 tests and a genuine tiny-GRU CPU training run independently confirmed by the user for v0.6. The user has not yet reported running the v0.7 HF preflight or real PEFT training.

What is complete

• Phases 1-3: configuration, metric detection and durable simulated attempt recovery.

• Phase 4: mock Meta agent produces typed, policy-checked advice; live hosted API is unverified.

• Phase 5A: real PyTorch CPU model training, loadable optimizer/model checkpoint and hash verification.

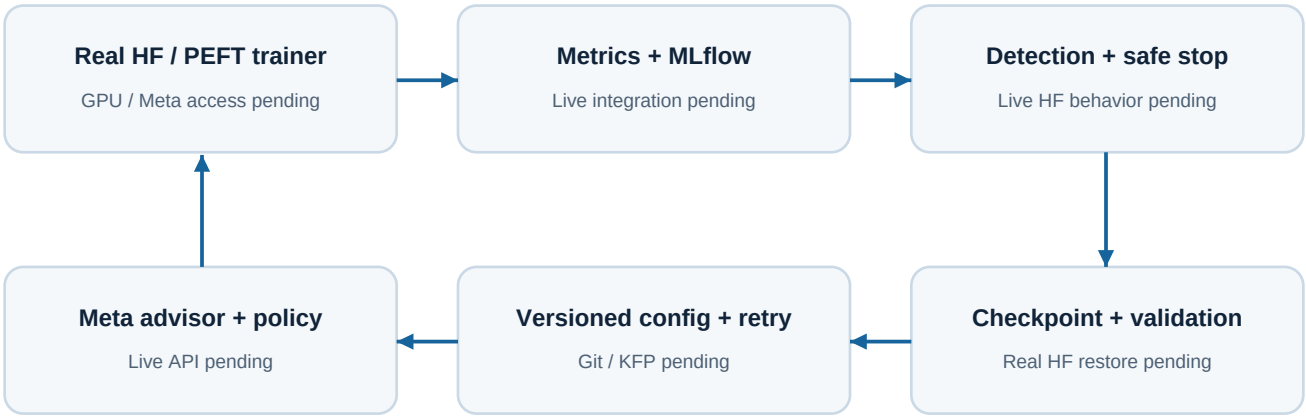
• Phase 5B: HF/PEFT/LoRA source and preflight contracts tested offline; no real LoRA/QLoRA GPU training yet.

What remains

Stage	Deliverable and exit gate
5B / live agent	Real LoRA adapter train, reload and full optimizer resume; live Meta advice validated
6 / MLflow	Durable metrics, parameters, datasets, decisions, checkpoints and lineage
7 / Kubeflow	Running Kubernetes GPU jobs, storage and recovery reconciliation
8 / HPO	Katib trial search, bounded early stop; Ray Tune only as an alternative
9 / Git edits	Versioned, tested configuration patches and rollback / approval
10 / UI	Live training monitoring, trials, actions, GPU usage and cost
11-12 / release	Holdout gates, registry, deployment, hardening, end-to-end chaos testing

System Integration Still to Prove

The target architecture is a bounded feedback loop; verification must show actual model weights and optimizer state, not simulated JSON metadata.



Target feedback loop; not yet demonstrated with real Meta Llama

Sequence for the next execution window

1. Run Phase 5B offline preflight on Windows and validate dataset/model identity, hyperparameters and evaluation schedule.

2. Execute a small public-model PEFT smoke test; validate saved adapter inference and a full Trainer resume.

3. Prepare licensed Meta Llama access and compatible CUDA or ROCm GPU stack; pin model revision and dependencies.

4. Capture aligned train/eval metrics and connect actual HF events to the stop-and-checkpoint controller.

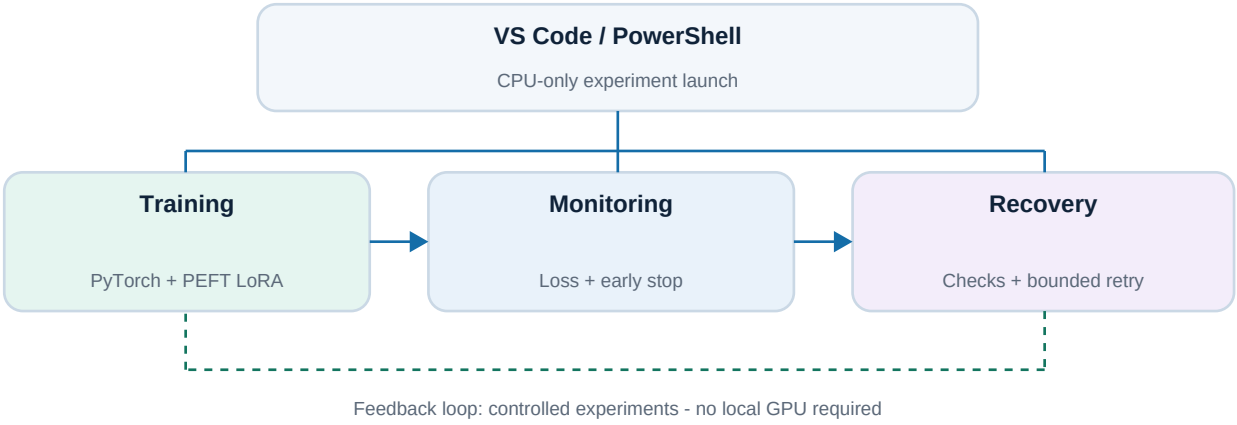
5. Implement MLflow persistence and a real model comparison before enabling remote Kubeflow retries.

Do not declare completion merely because tests pass. Required final evidence: genuine model anomaly, safe cancellation, verified checkpoint, policy-constrained advice, bounded restarted trial, independent validation and an audit trail.

Phase 5B: GPU-free execution path

Decision: the user has no dedicated GPU. The project can continue on native Windows using CPU-only PyTorch and small public Hugging Face models. Large Meta Llama/QLoRA GPU experiments are optional, not prerequisites for core agentic MLOps development.

Revised system architecture



Operational layers

Layer	CPU-first approach	Current verification
Training	PyTorch GRU verified; tiny GPT-2 PEFT next	GRU: Windows verified; PEFT: pending
Metrics / recovery	Aligned train/validation metrics, checkpoint hashes	Offline and CPU GRU contracts tested
Meta diagnostic AI	Hosted inference via API; local CPU makes requests	Mock tested; hosted call pending
MLflow	Local experiment server + artifacts	Planned
Kubeflow / Katib	Local CPU Kubernetes cluster if resources permit	Planned
GPU model training	Optional remote accelerator later	Not required for core MVP

Do not infer real LoRA quality from the tiny model. Existing 95 Windows PyTorch tests and 134 Phase 5B offline tests remain the historical evidence.

PHASE 5B | EXECUTION PLAYBOOK

Windows CPU verification and fine-tuning

From the project root, verify the existing offline plan without downloading any model:

```
.\.venv\Scripts\python.exe -m pytest -q
.\.venv\Scripts\python.exe -m autotune.lora_cli --config config/lora_public_cpu.yaml
```

For genuine LoRA training, first create a fresh environment. This preserves your working Phase 5A packages. An initial internet connection is required to install dependencies and download the small public model.

```
py -3.11 -m venv .venv-hf
.\.venv-hf\Scripts\python.exe -m pip install --upgrade pip
.\.venv-hf\Scripts\python.exe -m pip install torch
.\.venv-hf\Scripts\python.exe -m pip install -r requirements-lora.txt
```

Then execute the guarded CPU model run:

```
.\.venv-hf\Scripts\python.exe -m autotune.lora_cli --config config/lora_public_cpu.yaml --execute
--allow-cpu --output outputs/public-cpu-01
```

If the run succeeds, verify that its final LoRA adapter reloads for inference:

```
.\.venv-hf\Scripts\python.exe -m autotune.lora_cli --config config/lora_public_cpu.yaml --verify-adapter
outputs/public-cpu-01/final_adapter --allow-cpu
```

Inspect these output artifacts

metrics.jsonl and **monitor_decisions.jsonl** - recorded evaluation decisions; **checkpoints/** - full Hugging Face Trainer state; **final_adapter/** - adapter inference files; **result_receipt.json** - best validation result. These files will appear only after successful execution.

Safety restrictions

Use **hf.device: cpu** and **hf.quantization: none**. Do not use 4-bit QLoRA in the CPU pilot. **--allow-cpu** is explicit by design. An adapter-only artifact cannot resume optimizer and scheduler state. Use a separate fresh output directory for another attempt.

What constitutes progress without GPU

The full agentic MLOps loop remains the target: real training -> windowed metrics -> anomaly diagnosis -> safe stop -> validated parameter suggestion -> versioned configuration -> bounded retry -> independent validation comparison -> reviewable best artifact.

CPU-first revised milestones

1. LoRA CPU smoke - Load small public model, train adapters, obtain eval metrics, verify manifest and fresh adapter inference. [NOT YET LIVE-VERIFIED]

2. Full checkpoint resume - Interrupt at a known checkpoint, verify optimizer/scheduler/RNG and resume with matching provenance. [NOT YET LIVE-VERIFIED]

3. Durable self-healing integration - Connect HF Trainer stop and real artifacts to the existing policy agent; execute a controlled second CPU attempt. [NOT YET IMPLEMENTED]

4. MLflow observability - Track real CPU experiments, datasets, parameters, checkpoints and decision history. [PLANNED]

5. Kubernetes + Kubeflow/Katib - Run small resource-constrained CPU jobs and independent optimization trials on a local cluster. [PLANNED]

6. Optional GPU work - Scale to licensed Meta Llama fine-tuning on compatible remote accelerators when available. [OPTIONAL]

Current verification record

Phase 5A: 95/95 tests passed on the user's Windows machine and an actual tiny-GRU PyTorch checkpoint was saved. Phase 5B v0.7: 134/134 offline tests passed in the build environment. This v0.7.2 CPU-first update adds explicit CPU mode and a CLI guidance regression test; no live hosted Meta call, LoRA CPU run or GPU training is asserted.

Next checkpoint to report

Windows terminal output from **config/lora_public_cpu.yaml** preflight, optional real **--execute --allow-cpu** training, and adapter verification. Report evaluation loss and errors; do not paste access tokens.